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June 1, 2011

Mr. Lawrence Tomec, Director of Business Services
Brecksville Broadview Heights Board of Education
6812 Mill Road
Brecksville, OH 44141-1810

Re: Masonry and Floor Investigation
High School
Brecksville Broadview Heights School District
IAL No. 10496

Dear Mr. Tomec:

Please find enclosed with this letter, three copies of the investigation report for the slab settlement and masonry cracks at the south wing of the Brecksville Broadview Heights Consolidated School District. We trust that you will find the report satisfactory for your purposes.

As mentioned in the report, the next sequence of steps would be:

1. Perform invasive tests at various locations, such as interior walls, column enclosures, and exterior walls. This will allow us to verify how the masonry was constructed, what components were actually used and based on the evidence found the conditions at similar locations. Estimated engineering fee for this, including contractor's costs for performing the sampling, is estimated to be $ 5,000.00.
2. Perform additional geotechnical testing within the classrooms to obtain actual samples of the fill soils for laboratory testing. The estimated fees for the geotechnical testing is $ 8,000.00.
   Prepare preliminary plans of the masonry repairs, and the two alternate slab repairs. These would be sufficient to obtain budget estimates from qualified contractors. At this stage we would determine a phasing scheme so we can stay within the District's financing. The estimated fee for the preliminary drawings is $ 10,000.
3. Once the total scope of the work is developed and approved, a fee for preparing construction documents and providing construction administration will be offered.

Thank you for this opportunity to be of service. Please feel free to contact us with any questions or to request a formal proposal for engineering services described above..

Sincerely,
I A LEWIN, PE AND ASSOCIATES

Isaac Lewin, PE

file

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Brecksville, OH 44141-1810

Re: Masonry and Floor Investigation
High School
Brecksville Broadview Heights School District
IAL No. 10496

Dear Mr. Tomec:

As authorized by you, Isaac Lewin PE has reviewed the masonry and floor conditions at the high school for the Brecksville Broadview Heights School District Campus in Brecksville, Ohio. The review was requested due to the continued movement of the floor slab and cracking of the masonry. The school was visited on two occasions, to observe the existing conditions. The first visit was on November 10, 2010 and the second visit was on January 27, 2011. Both site visits were limited to visual observations of existing conditions. No testing or demolition was requested, authorized or performed as part of this site visit. Photographs of the existing conditions were taken and are on file at this office.

The investigation, in this writer's professional opinion, found there are two issues present in Units "D" and "E". One is the movement of the building beyond the allowances detailed into the building. It is the movement with no place to go, either through gaps in the masonry or lack of control joints, that is believed to be causing the majority of the masonry cracking. The second is the settlement of the floor slabs supported on soils. The fill used to construct the building appears not to have been compacted properly and is considered soft. The slab settlement also probably influences the cracking of the masonry on the first floor since the masonry column enclosures are supported on the slab rather than being supported on the column footing. The issues observed do not appear to affect the structural safety of the building, since the main structural framing elements are the steel frame, not the masonry. Further testing is needed to determine if the settlements are competed or if the soils will continue to consolidate and settle. Once this is known a decision can be made as to whether replacing the slabs with structural slabs supported on mini-piles or continue pressure grouting until the settlements stop.

The following documents were used in preparation of this report:

- Fanning Howey Drawings RA 1.13, RA 1.14, RA 1.19, RA 1.20, RA 1.27, RA 1.28, RA 1.29, S 1.7, and RG 2.2
- Professional Service industries "Report of Geotechnical Exploration Phase I", # 142-55027, dated April 10, 21995,
- Millisor & Nobil letter dated May 11, 2007
- Atlas Piers NEO letter dated May 14, 2007
- Atlas Piers NEO letter dated August 23, 2007
- Atlas Piers NEO letter dated January 16, 2008
- Atlas Piers NEO letter dated 2, 2008
- Timmerman Geotechnical Group Inc. "Floor Evaluation Report" TGG #067059A, dated July 10, 2008

The existing high school was designed and constructed between 1995 and 1997. The building was designed by Fanning Howey Associates Inc. of Columbus, Ohio and Geary Moore and Ahrens, Inc of Brecksville, Ohio. The area of the building that is the subject of this examination is known as Unit "D" and "E" per the construction documents. For purposes of this report the orientation of the long axis of the building is north-south with the gymnasium is considered the north end of the building and the auditorium is the south end. Based on this Units "D" and "E" are considered the middle south of the building. This part of the building is a two story structure with a partial basement on the west side of the building. The foundation wall is a substantial cast in place cantilever style concrete retaining wall. The building uses a superstructure of steel framing with moment connections for lateral bracing. The floors are concrete deck on steel joists and beams. The roof is steel deck on steel joists and beams. The structural steel is surrounded by masonry construction. The masonry surrounds are supported on the floor slabs. The corridor walls and some classroom divider walls are masonry also. Most of the classroom partition walls are of gypsum wallboard construction. The exterior walls consist of an outer wythe of brick, air space and an in wythe of concrete masonry. The building's foundations are shallow conventional reinforced concrete footings. It was noted the site was raised from 5 feet to 15 feet with compacted fill. The ground floor slabs are cast in place concrete placed on the subgrade. This is as recommended by the PSI report.

Based on the correspondence, the slab movement issues and masonry cracking developed about 8 years after completion of the building. According to the Atlas Pier correspondence the slab problems began appearing in Rooms 165 through 167, which is in the Unit "E" portion of the building. Atlas piers installed a series of "jack" piles under the perimeter foundation wall and the interior column footings. Atlas Piers also injected a slurry material under the floors in an attempt to level the floors and prevent further settlement of the slabs. Based on the correspondence available, no work has been performed since 2008.

The Timmerman Geotechnical report found the soils under the floors to be soft soils. They performed one test in Section "D" in Room 144 and three tests in Unit "E", Rooms 165, 166 and 175. They consistently found depths of fill commensurate with the grading plan, RG 2.2. However, they found the soils to be soft for most of the fill depth. This is contrary to the requirement of the PSI report. Per Detail 14, Drawing S 1.7, a different effort and fill was used at column and wall footings, than the general fill used under the floors.

This writer performed a visual survey of the damages on January 27, 2011. The survey started at the north end of Unit "D". The first item noted is a masonry crack at the expansion joint construction at the junction of Units "C" and "D". Numerous hairline cracks, and drywall cracks were observed in the second floor east rooms. For Unit "D" a total of 6 cracks of concern were observed. In Unit "E", the same cracking conditions were observed. These were evenly located throughout this Unit. A total of seventeen cracks of both types were observed in this Unit. The same issues were observed on the first floor, including drywall cracking and masonry cracking. In addition, locations where the slab is either settling, lifting and cracking were observed in both Units. In room 147, the slab had settled at least 1/2 inch at the outside wall. In Room 154 the floor appears to have raised. In Rooms 155 and 156, the slab joint has widened sufficiently to tear the carpeting. Unit "E" has similar issues to Unit "D", except the floor movement in Rooms 164/165, 166/167 and 174 are much more significant. Please note that these rooms had "jack" piles and underslab mudjacking installed. Movement of the masonry at the entrance to the Auditorium was observed.

Based on these observations there is a correlation between the slab settlements and the presence of fill (as referenced in the Timmerman report) and some correlation with the amount of masonry cracking on the second floor. The Timmerman report found the soils, for essentially the full depth of the fill, to be inadequately compacted. The worst areas for masonry cracking is in Unit "E", but the masonry cracking is present in significant amounts in Unit "D". In reviewing the problems, no signs of building settlement of the exterior walls or at interior columns were observed. Also the second floor was level based on walking the floor. The only indications of soil issues are the slab settlements in both Units "D" and "E". The exterior brick veneer shows no signs of unusual cracks or settlement.

The construction details were reviewed after the issues observed were correlated with location. The masonry construction is strictly used for closing the building in and are intended to provide structural stability to the building. The building's steel framing is designed as ordinary steel moment frame. This causes the steel to be continuous through each section between expansion joints and any movement in the steel is transmitted to the remaining portions of the units. The expansion joint for the south half of the building is located at the junction of Units "C" and "D". It was also observed that there are no masonry control joints present or detailed on the plans for the exterior walls for Units "D" and "E". The control joints will aid in preventing masonry issues due to building movement. Any movement will accumulate along the length of the building. It was also noted the masonry surrounding the columns were designed to be tight to the steel columns, therefore should the column move, the column will immediately push on the masonry, causing cracks to occur.

Based on this review of the existing conditions, design drawings, past investigations, it is this writer's professional opinion there are two issues present that are not related for the most part. The first issue is the interior surface cracking of the masonry, at both interior and exterior wall locations. It is this writer's professional opinion the building was constructed with the masonry too tight to the building framing. When the building's frame moves, due to temperature changes or loads, the steel impinges on the masonry and with sufficient movement occurs to crack the masonry and adjacent drywall. The second issue is the settlement of the slabs. Although the building foundations do not appear to be settling, the slabs definitely are. This has been ongoing problem since 1997, when the school opened. The past attempts to mudjack (pressure grout) the voids under the slabs have

failed since the slabs continue to settle. Since some of the settlements are up to 2 inches this amount is disconcerting and renders some rooms (164/167) potentially unusable. The settlement of the slabs has also aggravated the cracking in the masonry on the first floor since the masonry is supported on the floor and should the floor move the masonry will move and crack. Based on the evidence observed and the analysis presented, there appears no structural safety issues with the building.
To correct these problems, the following suggestions are offered:

Masonry Cracks

1. Perform an investigation into the connections, both quantity and quality, between the brick veneer and CMU backup.
2. Verify, by removal of masonry, that no joint reinforcing is tying the masonry together across any of the control joints.
3. Detail the installation of new control joints in the masonry walls, that would include the brick veneer and concrete masonry as well as the interior walls. Current industry standards call for such control joints to be spaced 20 feet on center.
4. Determine a method to isolate the masonry surrounds from being slab supported.

Slab Settlements

1. Perform additional testing of the soils below the slabs to determine the type of soils present, with sufficient samples taken to determine if the soils will continue to consolidate and cause additional slab settlement.
2. Decide, with that information available, if the slabs should be replaced with reinforced structural concrete slabs supported by mini-piles or if a program of pressure grouting the slabs on a regular basis should be instituted. Replacement of the slabs with a structural system will cause a lengthy shutdown of the rooms and construction issues for a building in use. However, this would be a onetime expense as opposed to the pressure grouting the slabs on an annual or multiple year basis.
3. The structural slabs will need to be isolated from the building columns to avoid adding any loads from the ground floor to the columns so as to not to overload them.

This office is able to assist you in obtaining the testing recommended and determining the costs and details needed to make the corrections noted.

Attached to this letter, are copies of Drawings RA 1.13, RA 1.14, RA 1.19 and RA 1.20, copies of details taken from drawings RA 1.27, RA 1.28, and RA 1.29 showing the details of construction for the original construction, selected photographs of existing conditions.

This letter is based on a limited examination of an existing, finished structure. No warranty is made or implied that all defects were observed and reported. Due to the finished construction, concealed conditions may exist that would affect this report; this office assumes no liability for concealed conditions that may affect the extent or costs of any repair work needed. If conditions assumed by this office are found to be different than those described in this letter, this office should be notified. This office reserves the right to adjust the information presented in this letter should additional information become available. This office does not assume any responsibility for any defects in design or construction, whether observed or not, since this office was not involved in the original design and construction of this building. The opinions, conclusions and recommendations contained in this letter are based on this writer's judgment and experience as a practicing Structural Engineer.

Please feel free to contact us with any questions or if we can be of further assistance in resolving these issues.

Sincerely,
I A LEWIN, PE AND ASSOCIATES

Isaac Lewin, PE

STATE OF OHIO
ISAAC
A
LEWIN
E-044897
REGISTERED
PROFESSIONAL ENGINEER

file
attachments